

The Department of Agriculture and Water Resources has completed its [group pest risk analysis \(PRA\)](#) for thrips and orthotospoviruses on fresh fruit, vegetable, cut-flower and foliage imports.

As our first group PRA, we considered the biosecurity risk posed by plant-feeding thrips insects (from the insect order Thysanoptera) that are, or are likely to be, associated with fresh fruit, vegetables, cut flowers and foliage imported into Australia as commercial consignments. We also assessed the emerging risks posed by all members of the virus genus Orthotospovirus (formerly ‘tospovirus’) that are transmitted by some thrips.

This group PRA did not address the risks posed by thrips or orthotospoviruses on nursery stock imports because these will be considered in a separate review. We will consult with stakeholders if any changes are made to existing nursery stock import conditions.

We followed 3 principal steps when conducting this group PRA:

- Our experts conducted a review of scientific knowledge relevant to the group of pests and prepared a PRA.
- We released the draft group PRA for a 90-day public consultation period on 14 December 2016 via [Biosecurity Advice 2016-35](#). The draft report identified the biosecurity risks and risk management measures required to achieve Australia’s appropriate level of protection (ALOP).
- We finalised the group PRA, taking into consideration all stakeholder submissions, and released the [final group pest risk analysis for thrips and orthotospoviruses on fresh fruit, vegetable, cut-flower and foliage imports](#) on 30 November 2017 via [Biosecurity Advice 2017-26](#).

This group PRA is funded through the Australian Government’s [Agricultural Competitiveness White Paper](#) to strengthen biosecurity surveillance and analysis.

Final Report Summary

The Australian Government Department of Agriculture and Water Resources is improving the effectiveness and consistency of the PRA process. A key step in this improvement is the development of the group PRA, which considers the biosecurity risk posed by groups of pests across numerous import pathways. It applies the significant body of available scientific knowledge, including pest interception data and previous PRAs, to provide an overarching analysis of the risks posed by the group.

The International Plant Protection Convention (IPPC) defines PRA as ‘the process of evaluating biological or other scientific and economic evidence to determine whether an organism is a pest, whether it should be regulated, and the strength of any phytosanitary measures to be taken against it’ (FAO 2016b). International Standard for Phytosanitary Measures (ISPM) 2: Framework for pest risk analysis (FAO 2016a), states that ‘Specific organisms may ... be analysed individually, or in groups where individual species share common biological characteristics.’ This is the basis for the group PRA, in which organisms are grouped if they share common biological characteristics, and as a result also have similar likelihoods of entry, establishment and spread and comparable consequences—thus posing a similar level of biosecurity risk.

Undertaking and utilising PRAs on groups of pests that share common biological characteristics provides significant opportunities to improve effectiveness and consistency of commodity-based PRAs with which those pests are also associated and to maintain a high level of biosecurity protection against new and emerging risks. The group approach to PRA was initiated by the department to take advantage of these opportunities. Each group PRA is a 'building block' that can be used to review existing trade pathways, and can also be applied to prospective pathways for which a specific PRA is required.

If a group PRA is used to review existing or new trade pathways there may be no need to undertake further detailed PRAs on these pests—if the trade-dependent factors relating to the likelihood of entry on specific pathways have been verified, the group PRA can be applied.

This is the first group PRA to be finalised—further group PRAs are underway. This group PRA considers the biosecurity risk posed by all members of the insect order Thysanoptera (commonly referred to as thrips) and all members of the virus genus Orthotospovirus (formerly tospovirus) that are (or are likely to be) associated with fresh fruit, vegetables, cut-flowers or foliage imported into Australia as commercial consignments. It also assesses the emerging risks posed by orthotospoviruses, which are transmitted by some thrips.

The genus tospovirus has recently undergone taxonomic revision by the International Committee on Taxonomy of Viruses (ICTV 2017) being renamed Orthotospovirus and assigned to the new family Tospoviridae and new order Bunyavirales. This revision will be applied in this report to all 30 species formerly described as tospoviruses, as appropriate.

Thrips and the orthotospoviruses they transmit can cause considerable economic consequences across a wide range of fruit, vegetable, legume and ornamental crops by reducing yield, quality and marketability. Orthotospoviruses are a significant emerging risk to Australia with many recent reports of new species with rapidly expanding host plant ranges, geographic distributions and thrips vectors.

This group PRA identifies and analyses the key quarantine pests of biosecurity importance to Australia in these two groups of organisms. It is built on a foundation of 18 years of PRAs undertaken by the department—all of which were subjected to robust scientific analyses and extensive processes of stakeholder consultation. These pest risk assessments showed marked consistency in the level of biosecurity risk posed by thrips relative to the appropriate level of protection (ALOP) for Australia. They also indicated that certain thrips species are associated with a broad range of plant commodities from many countries.

This report's conclusions have been validated with available scientific evidence including 26 years of interception data collected at Australia's borders, similar interception records available from other countries and an extensive literature review. The report includes significant pests that have been recognised internationally, or by Australian industry, or those identified by states and territories as regional pests for Australia.

This report does not address the risk posed by thrips and orthotospoviruses on nursery-stock imports, which are another significant commercial pathway for the possible introduction of these pests. These

will be considered in a separate review. The department will consult with stakeholders if any changes are made to existing nursery-stock import conditions.

The order Thysanoptera comprises more than 6,000 described thrips species within nine families. This group PRA identified the thrips families that are not likely to be associated with fresh fruit, vegetable, cut-flower and foliage imports, or have no potential for economic consequences for Australia and cannot meet the definition of a quarantine pest. As a result, only the phytophagous (plant-feeding) members of the Thripidae and phytophagous members of the Phlaeothripidae were identified as potential quarantine pests for Australia. These phytophagous thrips are the focus of this group PRA.

Selection criteria were used to identify thrips species within the identified phytophagous Thripidae and the phytophagous Phlaeothripidae with potential biosecurity importance for Australia. Within this group, 79 thrips species were confirmed as quarantine pests for Australia. The final group PRA also identified 27 orthotospoviruses that are quarantine pests for Australia.

These thrips and orthotospovirus quarantine pests were all estimated to have an ‘indicative’ unrestricted risk estimate of ‘low’, which does not achieve the ALOP for Australia. These risk estimates are ‘indicative’ because the likelihood of entry for quarantine pests can be influenced by a range of factors relating to specific trade pathways.

Fourteen thrips species are known to naturally transmit orthotospoviruses. Eleven of these are already regarded as quarantine pests for Australia. The remaining three are present in Australia and not under official control. This group PRA recommends that the regulatory status of these three thrips species—*Frankliniella schultzei*, *Scirtothrips dorsalis* and *Thrips tabaci*—be changed from non-regulated to regulated because these thrips can carry and transmit quarantine orthotospoviruses. This change is not expected to significantly affect trade.

Initial evaluation of six viruses other than orthotospoviruses that are transmitted by thrips was also undertaken in this group PRA. The department will undertake further separate analysis for Maize chlorotic mottle virus and has sought further information on viruses of potential regional concern to Western Australia (Sowbane mosaic virus, Tobacco streak virus and Strawberry necrotic shock virus). The thrips vector of Pelargonium flower break virus is regulated, which also mitigates the risk from this virus. Prunus necrotic ringspot virus is not a quarantine pest.

Phytosanitary measures are identified in this final report for use in specific cases where measures are required. These measures are consistent with long-standing established policy for quarantine thrips and also mitigate the risk posed by the quarantine orthotospoviruses they transmit.

Imported commodities will be regulated if they are infested with quarantine pest thrips or regulated thrips that transmit quarantine orthotospoviruses to reduce the risk of establishment of these organisms in Australia. Regulation will be in accordance with the final group PRA and any other relevant commodity-based PRAs.

The final group PRA identifies measures for quarantine and regulated thrips and alternative risk management options that may be considered on a case-by-case basis when developing new import

conditions for specific commodities, or reviewing existing import conditions for commodities that are currently traded.

Where measures are required, they will include:

- freedom from quarantine and regulated thrips and
- verification, such as inspection, to provide assurance that Australia's import conditions have been met and appropriate level of protection achieved.

Imported goods that are frequently found to be infested with thrips may be subject to mandatory treatment.

Written submissions on the draft report were received from five stakeholders. The final report takes into account stakeholder comments on the draft report. The department has made a number of changes to this group PRA following consideration of these comments, and additional review of the literature. These changes include:

- Explaining further the basis for assessing phytophagous thrips as a group, including that they 'share common biological characteristics', a term used in the International Standards for Phytosanitary Measures
- Renaming and revising Chapter 2 to add additional text on thrips biology
- Adding additional evidence to support the removal of Capsicum chlorosis virus–Phalaenopsis strain as a quarantine pest for Australia
- Revising the likelihood of spread for orthospoviruses from Moderate to High
- Rewording text to provide more clarity for reasoning and conclusions.

Responses to key issues raised by stakeholders are presented in Appendix I.

Next steps

We will use this group PRA when reviewing existing import conditions, or when developing new import conditions, for specific commodities when thrips are identified.

New scientific information

Scientific information can be provided to us at any time, including after a risk analysis has been completed. We will consider the information provided and review the analysis.

More information

For more information, you can [email us](#) or phone +61 2 6272 5094.